Trials@uspto.gov
571.272.7822

Paper 51
Entered: April 29, 2025

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

BEONE MEDICINES USA, INC. and BEIGENE, LTD.,
Petitioner,

v.

PHARMACYCLICS LLC,
Patent Owner.

PGR2024-00009
Patent 11,672,803 B2

Before SHERIDAN K. SNEDDEN, SUSAN L. C. MITCHELL, and TINA E. HULSE, *Administrative Patent Judges.*

PER CURIAM

JUDGMENT
Corrected Final Written Decision
Determining All Challenged Claims Unpatentable
35 U.S.C. § 328(a)

I. INTRODUCTION

BeOne Medicines USA, Inc.[1] and Beigene, Ltd. (collectively, "Petitioner") filed a Petition requesting post-grant review of claims 1–8, 12, and 20 ("the Challenged Claims") of U.S. Patent No. 11,672,803 B2 (Ex. 1001, "the '803 Patent"), which is owned by Pharmacyclics LLC ("Patent Owner"). Paper 2 ("Pet."). After considering the Petition, Preliminary Response (Paper 6), Petitioner's pre-institution Reply (Paper 7), and Patent Owner's pre-institution Sur-Reply (Paper 8), we instituted post-grant review of the Challenged Claims of the '803 Patent. Paper 9 ("Institution Decision" or "Dec. Inst.").

After institution, Patent Owner filed a Response (Paper 23, "PO Resp."), Petitioner filed a Reply (Paper 29, "Pet. Reply"), and Patent Owner filed a Sur-reply (Paper 36, "PO Sur-reply"). An oral argument was held in this proceeding on January 30, 2025, and a copy of the transcript was entered into the record. Paper 47 ("Tr.").

We have jurisdiction under 35 U.S.C. § 6, and we issue this Final Written Decision under 35 U.S.C. § 328(a) and 37 C.F.R. § 42.73. For the reasons discussed below, we conclude that Petitioner has proven by a preponderance of the evidence that claims 1–8, 12, and 20 of the '803 Patent are unpatentable.

---

[1] BeOne Medicines USA, Inc. was formally known as BeiGene USA, Inc. at the time the Petition was filed. Pet. 3. Petitioner notified the Board on March 20, 2025, that BeiGene USA, Inc. changed its name to BeOne Medicines USA, Inc. via a Certificate of Amendment of Incorporation filed with the Secretary of State of Delaware. Paper 49.

II. BACKGROUND

A. Real Parties-in-Interest

Petitioner identifies itself as the real parties-in-interest. Pet. 3; Paper 49.

Patent Owner identifies itself as a real party-in-interest and states that it is a wholly owned subsidiary of AbbVie Inc. and the sole assignee of the ’803 Patent. Paper 4, 1. Patent Owner also states that Janssen Biotech, Inc. is the exclusive licensee of the ’803 Patent. *Id.*

B. Related Proceedings

Petitioner and Patent Owner identify *Pharmacyclics LLC v. BeiGene USA, Inc.*, Case No. 1:23-cv-00646 (CFC) (D. Del.) filed on June 13, 2023, as a related matter.[2] Pet. 4; Paper 4, 1. Patent Owner also states that U.S. Patent Application Nos. 18/180,064 and 18/171,254, which are pending, claim the benefit of the application that led to the ’803 Patent. Paper 4, 1.

C. The ’803 Patent

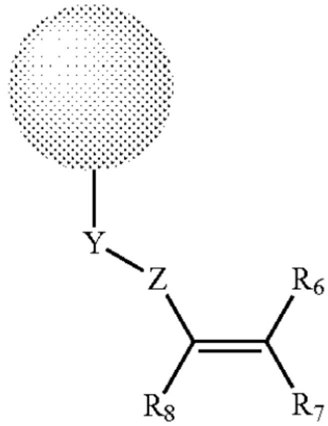
The ’803 Patent, entitled “Use of Inhibitors of Brutons Tyrosine Kinase (BTK),” issued from U.S. Patent Application No. 16/926,280 (“the ’280 Application”), which was filed July 10, 2020. Ex. 1001, codes (54), (21), (22). The ’803 Patent claims the benefit of the filing dates of a series of applications, the earliest of which is U.S. Patent Application No. 13/153,317 (“the ’317 Application”), filed on June 3, 2011. *Id.*, code (63). The ’803 Patent also claims the benefit of the filing dates of

[2] The district court proceeding has been stayed until resolution of this post-grant proceeding. Pet. 55.

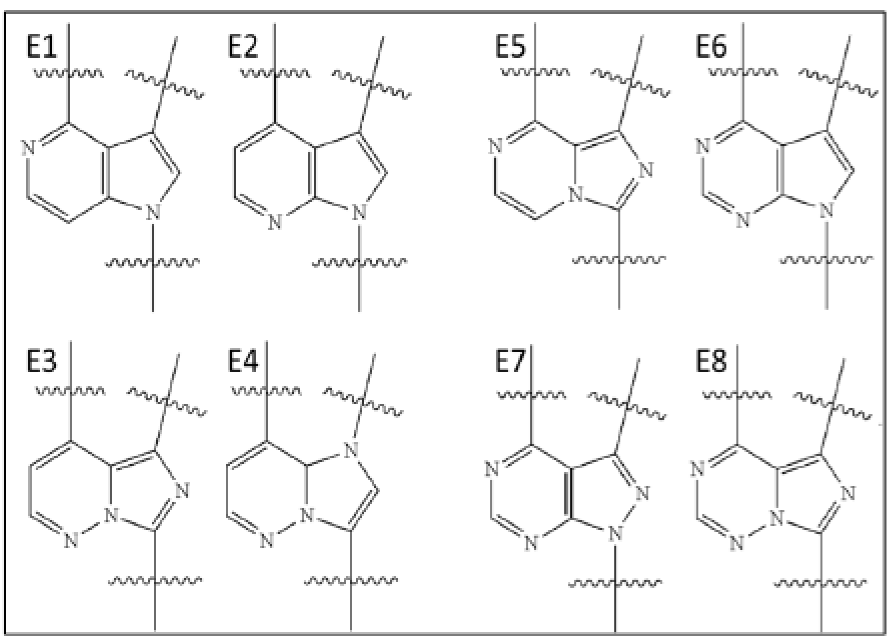
various provisional applications, the earliest of which was filed on June 3, 2010. *Id.*, code (60).

The ’803 Patent relates to methods for treating a cancer, such as a hematological malignancy, by administering an irreversible Bruton’s tyrosine kinase (“BTK”) inhibitor (“BTKI”) until disease progression, unacceptable toxicity, or individual choice. Ex. 1001, Abstract, 1:37–2:1, 3:24–46. The ’803 Patent explains that BTK is a key signaling enzyme expressed in all hematopoietic cell types except T lymphocytes and natural killer cells. *Id.* at 1:37–40. BTK is a key regulator of B-cell development, activation, signaling, and survival and plays a role in a number of other hematopoietic cell signaling pathways. *Id.* at 1:40–59. According to the ’803 Patent, BTKIs can induce mobilization of lymphoid cells (or, in some cases, cause lymphocytosis, i.e., an increase in lymphocytes) in solid hematological malignancies, which increases their exposure to additional cancer treatments and their availability for biomarker screening. *Id.* at 11:16–22.

The ’803 Patent states that in some embodiments, BTKIs form an irreversible covalent bond with a cysteine sidechain of BTK, a BTK homolog, or a BTK tyrosine kinase cysteine homolog. Ex. 1001, 3:18–21. The BTKIs described in the ’803 Patent “are selective for Btk and kinases having a cysteine residue in an amino acid sequence position of the tyrosine kinase that is homologous to the amino acid sequence position of cysteine 481 in Btk.” *Id.* at 45:56–60. The ’803 Patent also discloses generic structures of various BTKIs, including Formulas (A)–(F), with numerous possible substituents in various positions. *Id.* at 47:5–80:36. For example, Formula (E) is reproduced below:



Formula (E) depicts the structure of an irreversible BTKI (i.e., an "iBTKI") with an orb that represents "a moiety that binds to the active site of a kinase, including a tyrosine kinase, further including a Btk kinase cysteine homolog." Ex. 1001, 62:15–17. According to the '803 Patent, "[i]n some embodiments, [the orb] is a substituted fused biaryl moiety selected from [eight different structures]." *Id.* at 62:44–63:20. Those eight structures (annotated as E1 to E8) are reproduced below:



Id. (annotated by Petitioner). The eight structures depicted are all 5:6 heterocyclic rings with two to four nitrogen heteroatoms.

The Specification also includes Examples 6, 10–12, 14, and 15, which describe clinical trials and/or other testing to determine the efficacy of a BTKI in chronic lymphocytic leukemia ("CLL") and small lymphocytic leukemia ("SLL") patients. *Id.*, Examples 6, 10–12, 14, 15; Ex. 1005 ¶¶ 118–135. For example, the Specification states in Example 6 that there was an initial rapid reduction in nodal disease and spleen size with a corresponding rise in the circulating lymphocyte count (i.e., lymphocytosis). Ex. 1001, 135:30–34. At the same time, patients reported symptomatic improvement, and over time, the initial rise in lymphocytes returns to pre-treatment levels. *Id.* at 135:39–41.

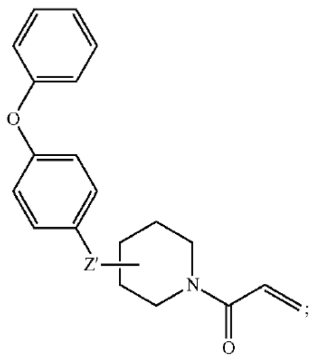
The Specification states that "[c]onsidering the age of the individuals typically diagnosed with CLL and SLL, there is a need in the art for a simple and effective treatment of the disease with minimum side-effects that do not impede on the patient's quality of life." *Id.* at 36:18–22. According to the '803 Patent, "[t]he instant invention fulfills this long standing need in the art." *Id.* at 36:22–23.

D. Illustrative Claims

Petitioner challenges claims 1–8, 12, and 20 of the '803 Patent, of which claims 1 and 8 are independent. Claims 1 and 8 are illustrative and are reproduced below:

> 1. A method for treating chronic lymphocytic leukemia (CLL) or small lymphocytic lymphoma (SLL) in an individual comprising:
>
> orally administering to the individual a therapeutically effective amount of an irreversible inhibitor of Bruton's tyrosine kinase (Btk) on a continuous daily regimen until progression of the CLL or SLL or unacceptable toxicity;
>
> wherein lymphocytosis is not considered progression of the CLL or SLL, and

wherein the irreversible inhibitor of Btk is a small organic molecule having the structure:



wherein Z′ is an optionally substituted fused heterocyclic ring system comprising from 2-4 nitrogen heteroatoms;

wherein the optionally substituted fused heterocyclic ring system consists [of] a 5-membered ring comprising at least one nitrogen heteroatom fused to a 6-membered ring comprising at least one nitrogen heteroatom; and

wherein the irreversible inhibitor of Btk forms a covalent bond with Cys^{481} of a Bruton's tyrosine kinase.

Ex. 1001, 151:34–152:4.

8. A method for treating chronic lymphocytic leukemia (CLL) or small lymphocytic lymphoma (SLL) in an individual comprising:

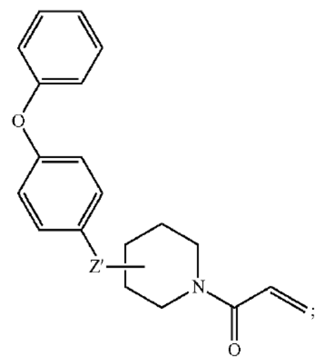
orally administering to the individual a therapeutically effective amount of an irreversible inhibitor of Bruton's tyrosine kinase (Btk) on a continuous daily regimen until progression of the CLL or SLL or unacceptable toxicity;

wherein:

lymphocytosis is not considered progression of the CLL or SLL;

said administration of the therapeutically effective amount is an amount that results in >90% of the Btk active sites in the peripheral blood mononuclear cells of the individual being occupied by the irreversible inhibitor of Btk twenty-four hours following said administration; and

the irreversible inhibitor of Btk is a small organic molecule having the structure:



> wherein Z′ is an optionally substituted fused heterocyclic ring system comprising from 2-4 nitrogen heteroatoms; and
> wherein the optionally substituted fused heterocyclic ring system consists [of] a 5-membered ring comprising at least one nitrogen heteroatom fused to a 6-membered ring comprising at least one nitrogen heteroatom.

Id. at 152:46–153:23.

The challenged dependent claims require a specific BTK occupancy (claim 2), using pharmacodynamic parameters to determine the therapeutically effective dose (claim 3), administering a fixed dose of BTKI (claim 4), achieving specific pharmacokinetic results from administering a therapeutically effective amount (claim 5), treating CLL and SLL with once daily oral administration of BTKI (claims 6 and 7), and having three nitrogen heteroatoms in the fused heterocyclic ring system (claims 12 and 20). *Id.* at 152:6–45 (claims 2–7), 154:20–21 (claim 12), 156:20–21 (claim 20).

E. The Asserted Grounds of Unpatentability

Petitioner challenges claims 1–8, 12, and 20 of the ’803 Patent based on the grounds set forth in the table below.

Claims Challenged	35 U.S.C. §	Reference(s)/Basis
1–8, 12, 20	112(a)	Written Description
1–8, 12, 20	112(a)	Enablement

Claims Challenged	35 U.S.C. §	Reference(s)/Basis
1–8, 12, 20	102(a)	Byrd[3]
1–8, 12, 20	103	Byrd and IMBRUVICA Label[4]

Petitioner relies on the Declarations of Hollis Showalter, Ph.D. (Exs. 1003, 1105), the Declaration of Steven Rosen, M.D. (Ex. 1005), and the Declaration of James M. Veal, Ph.D. (Ex. 1106).

Patent Owner relies on the Declaration of Michael J. Eck, M.D., Ph.D. (Ex. 2010), the Declaration of David A. Scott, Ph.D. (Ex. 2012), and the Declaration of Stephen E. Spurgeon, M.D. (Ex. 2014).

III. ELIGIBILITY FOR POST-GRANT REVIEW

Section 6(d) of the Leahy-Smith America Invents Act, Pub. L. No. 112-29, 125 Stat. 284 (Sept. 16, 2011) ("AIA") sets forth the post-grant review provisions, which apply only to patents subject to the first-inventor-to-file provisions of the AIA. AIA § 6(f)(2)(A) (stating the provisions of Section 6(d) "shall apply only to patents described in section 3(n)(1)"). Post-grant reviews are only available for patents that issue from applications "that contain[] or contained at any time . . . a claim to a claimed invention that has an effective filing date . . . on or after" March 16, 2013. AIA § 3(n)(1). Moreover, "[a] petition for a post-grant review may only be filed not later than the date that is 9 months after the date of the grant of the patent or of the issuance of a reissue patent (as the case may be)." 35 U.S.C. § 321(c). Petitioner has the burden of demonstrating eligibility for post-

[3] Byrd et al., *Targeting BTK with Ibrutinib in Relapsed Chronic Lymphocytic Leukemia*, 369 N. ENGL. J. MED. 32–42 (2013). Ex. 1009 ("Byrd").

[4] Highlights of Prescribing Information for IMBRUVICA® (ibrutinib) capsules, for oral use (2016). Ex. 1058 ("IMBRUVICA Label").

grant review. *See Mylan Pharms. Inc. v. Yeda Res. & Dev. Co.*, PGR2016-00010, Paper 9 at 10 (PTAB Aug. 15, 2016).

The ’803 Patent issued on June 13, 2023, and Petitioner filed the Petition on November 1, 2023. Ex. 1001, code (45). Thus, the Petition was timely filed less than nine months after the date the patent was granted.

The dispositive issue in this proceeding is whether each of the Challenged Claims of the ’803 Patent can claim priority to the earliest non-provisional patent application, the ’317 Application, which was filed on June 3, 2011. If they all can, then the ’803 Patent is not eligible for post-grant review. If any cannot, then the ’803 Patent is eligible for post-grant review. Because the specification of the ’317 Application is substantively identical to the Specification of the ’803 Patent, if the claims lack 35 U.S.C. § 112 support for priority purposes, then the claims lack § 112 support for unpatentability purposes, as well.

We, therefore, focus our analysis on the § 112 issue. But before we address that issue, we must first address the level of ordinary skill in the art and claim construction to determine the scope of the claims.

A. Person of Ordinary Skill in the Art

In determining the level of ordinary skill in the art, we consider the type of problems encountered in the art, the prior art solutions to those problems, the rapidity with which innovations are made, the sophistication of the technology, and the educational level of active workers in the field. *Custom Accessories, Inc. v. Jeffrey-Allan Indus., Inc.*, 807 F.2d 955, 962 (Fed. Cir. 1986).

Petitioner asserts that a person of ordinary skill in the art ("POSA") would have been:

> a team, with at least one member having at least a medical degree with training in hematology and oncology, including several years of clinical experience in treating hematologic malignancies, and at least another member having at least a Ph.D. in chemistry, organic chemistry, or related field with training as a medicinal chemist.

Pet. 57 (citing Ex. 1003 ¶ 136; Ex. 1005 ¶¶ 172–75).

Patent Owner agrees that defining a POSA as a team with a "clinician member" and a "chemist member" is appropriate given the subject matter of the claims. PO Resp. 27. Patent Owner asserts, however, that the POSA definition should expressly—rather than impliedly—account for knowledge of the structure-function relationship of kinase inhibitors. *Id.* at 28. Thus, Patent Owner would include in the definition "at least one member having at least a Ph.D. in chemistry, organic chemistry, biochemistry, or related field with training in the structure-function relationship of kinase inhibitors." *Id.* (citing Ex. 2010 ¶¶ 125–126). Moreover, Patent Owner contends that Petitioner's expert, Dr. Showalter testified that "'one of those members of the team[]' would be a 'specialist[] in molecular docking or in, say, x-ray crystallography or computational modeling[.]'" *Id.* (quoting Ex. 2018, 143:11–22). Nevertheless, Patent Owner states that even under Petitioner's definition, the challenges fail. *Id.*

Petitioner disagrees that a POSA must be an expert in computational modeling, noting that the Specification does not mention modeling and that medicinal chemists would not rely on modeling to screen the claimed compounds. Pet. Reply 3 (citing Ex. 1105 ¶¶ 10–12, 14; Ex. 1106 ¶¶ 33, 34, 36). According to Petitioner's expert Dr. Veal, modeling may be used for optimization efforts to "better understand the chemical interactions between the kinase inhibitor candidates and the target kinase complex in order to design analogs with improved potency." Ex. 1106 ¶ 34. But, in Dr. Veal's opinion, a POSA would never rely on modeling alone to determine whether a potential inhibitor would covalently bind BTK and would run assays instead. *Id.* ¶ 36.

Having considered the parties' respective arguments and evidence, we agree with Patent Owner that expressly including knowledge of the structure-function relationship of kinase inhibitors is appropriate here, given the subject matter of the claims. As for computational modeling, although the Specification may not expressly discuss modeling, we agree with Patent Owner that a POSA with knowledge of the structure-function relationship of kinase inhibitors would also be knowledgeable about computational modeling, particularly in light of the state of the art at that time. *See* Ex. 2010 ¶ 115 (citing Ex. 2035;[5] Ex. 2036;[6] Ex. 2037;[7] Ex. 2041[8]). Moreover, although he may disagree as to when the use of modeling is appropriate or useful, Dr. Veal admits that modeling may provide insight during the drug development process. Ex. 1106 ¶ 36.

---

[5] Friesner et al., *Glide: A New Approach for Rapid, Accurate Docking and Scoring. 1. Method and Assessment of Docking Accuracy*, 47 J. MED. CHEM. 1739–49 (2004) (Ex. 2035).
[6] Friesner et al., *Extra Precision Glide: Docking and Scoring Incorporating a Model of Hydrophobic Enclosure for Protein–Ligand Complexes*, 49 J. MED. CHEM. 6177–96 (2006) (Ex. 2036).
[7] Glide 5.5 User Manual (Schrodinger Press 2009) (Ex. 2037).
[8] Halgren et al., *Glide: A New Approach for Rapid, Accurate Docking and Scoring. 2. Enrichment Factors in Database Screening*, 47 J. MED. CHEM. 1750–59 (2004) (Ex. 2041).

Thus, we define a POSA as a member of a multidisciplinary team with at least one "clinician member" who has a medical degree with training in hematology and oncology, including several years of clinical experience in treating hematologic malignancies, and at least one "chemist member" who has a Ph.D. in chemistry, organic chemistry, or related field with training in the structure-function relationship of kinase inhibitors, including experience with computational modeling.

Regardless, we note that the experts testifying in this proceeding—Dr. Showalter, Dr. Rosen, Dr. Veal, Dr. Eck, Dr. Scott, and Dr. Spurgeon—are all qualified to opine from the perspective of a POSA, as each has at least ordinary skill in the art in their respective fields as either a clinician member or a chemist member of the POSA team. Ex. 1003 ¶¶ 4–11 (Showalter, chemist member); Ex. 1005 ¶¶ 7–27 (Rosen, clinician member); Ex. 1106 ¶¶ 6–16 (Veal, chemist member); Ex. 2010 ¶¶ 2–11 (Eck, chemist member[9]); Ex. 2012 ¶¶ 2–7 (Scott, chemist member); Ex. 2014 ¶¶ 6–17 (Spurgeon, clinician member); *see also Kyocera Senco Indus. Tools Inc. v. Int'l Trade Comm'n*, 22 F.4th 1369, 1376–77 (Fed. Cir. 2022) ("To offer expert testimony from the perspective of a skilled artisan in a patent case—like for claim construction, validity, or infringement—a witness must at least have ordinary skill in the art.").

[9] Although Dr. Eck has an M.D. as well as a Ph.D., it appears from his declaration and curriculum vitae that his primary focus is on medicinal chemistry rather than clinical medicine. Ex. 2010 ¶¶ 2–11. We, therefore, consider his testimony from the perspective of the chemist member of the POSA team.

B. Claim Construction

The Board applies the same claim construction standard that would be used to construe the claim in a civil action under 35 U.S.C. § 282(b). 37 C.F.R. § 42.200(b). Under that standard, claim terms "are generally given their ordinary and customary meaning" as understood by a person of ordinary skill in the art at the time of the invention. *Phillips v. AWH Corp.*, 415 F.3d 1303, 1312–13 (Fed. Cir. 2005) (en banc). Moreover, "claim terms need only be construed 'to the extent necessary to resolve the controversy.'" *See Wellman, Inc. v. Eastman Chem. Co.*, 642 F.3d 1355, 1361 (Fed. Cir. 2011) (quoting *Vivid Techs., Inc. v. Am. Sci. & Eng'g, Inc.*, 200 F.3d 795, 803 (Fed. Cir. 1999)).

The parties agree that the preamble of the claims should be limiting and that the terms "optionally substituted fused heterocyclic ring system" and "irreversible" BTKIs should be construed consistent with the definitions provided in the Specification. Pet. 42–45; *see also id.* at 42 (citing Ex. 1001, 19:57–20:7 (defining "optionally substituted" or "substituted"), 14:51–52 (defining "fused"), 15:66–16:21 (defining "heterocycle"), 14:36–44 (defining "ring" and "ring system"), 26:23–25 (defining "irreversible Btk inhibitor"); PO Resp. 26.

The parties also agree that the limitation "wherein lymphocytosis is not considered progression of the CLL or SLL" is entitled to patentable weight because it requires the physical action of continuing to administer the iBTKI after lymphocytosis. Pet. 47; PO Resp. 26; *see Praxair Distrib., Inc. v. Mallinckrodt Hosp. Prod. IP Ltd.*, 890 F.3d 1024, 1033 (Fed. Cir. 2018) ("[A] limitation that merely claims information by incorporating that information into a mental step will receive patentable weight only if the limitation is functionally related to the substrate.").

Finally, Petitioner argues that the preamble should be construed "in combination with the term 'therapeutically effective amount' to limit the scope of the Challenged Claims to the use of certain irreversible BTKIs that both (1) effectively treat CLL/SLL ('efficacy'), and (2) do so without causing undue adverse side effects ('safety')." Pet. 42 (citing Ex. 1005 ¶¶ 164–68); *see also id.* at 45–47. Petitioner contends that "if a particular irreversible BTKI cannot be administered to effectively and safely treat CLL/SLL at any amount, then administering that BTKI does not practice the claims." *Id.* at 46.

Patent Owner asserts that Petitioner's proposed construction of "therapeutically effective amount" conflicts with the express definition in the Specification. PO Resp. 26–27. That is, the Specification states that the term "therapeutically effective amount" refers to "a sufficient amount of an agent or a compound being administered which will relieve to some extent one or more of the symptoms [of a] B-cell lymphoproliferative disorder (BCLD) . . . [or] reduction and/or alleviation of the signs, symptoms, or causes of BCLD." Ex. 1001, 24:58–64.

Having considered the parties' respective arguments and evidence, we agree with the parties that the preamble is limiting as a statement of intentional purpose. *See Eli Lilly & Co. v. Teva Pharms. Int'l GmbH*, 8 F.4th 1331, 1342 (Fed. Cir. 2021) (stating the Federal Circuit "has not hesitated to hold preambles limiting when they state an intended purpose for methods of using a compound"). Moreover, we determine that each of the proposed claim terms, including "therapeutically effective amount," should be construed according to their express definitions in the Specification.[10] *See Phillips*, 415 F.3d at 1316 (stating "the inventor's lexicography governs" when the specification reveals a special definition for a claim term).

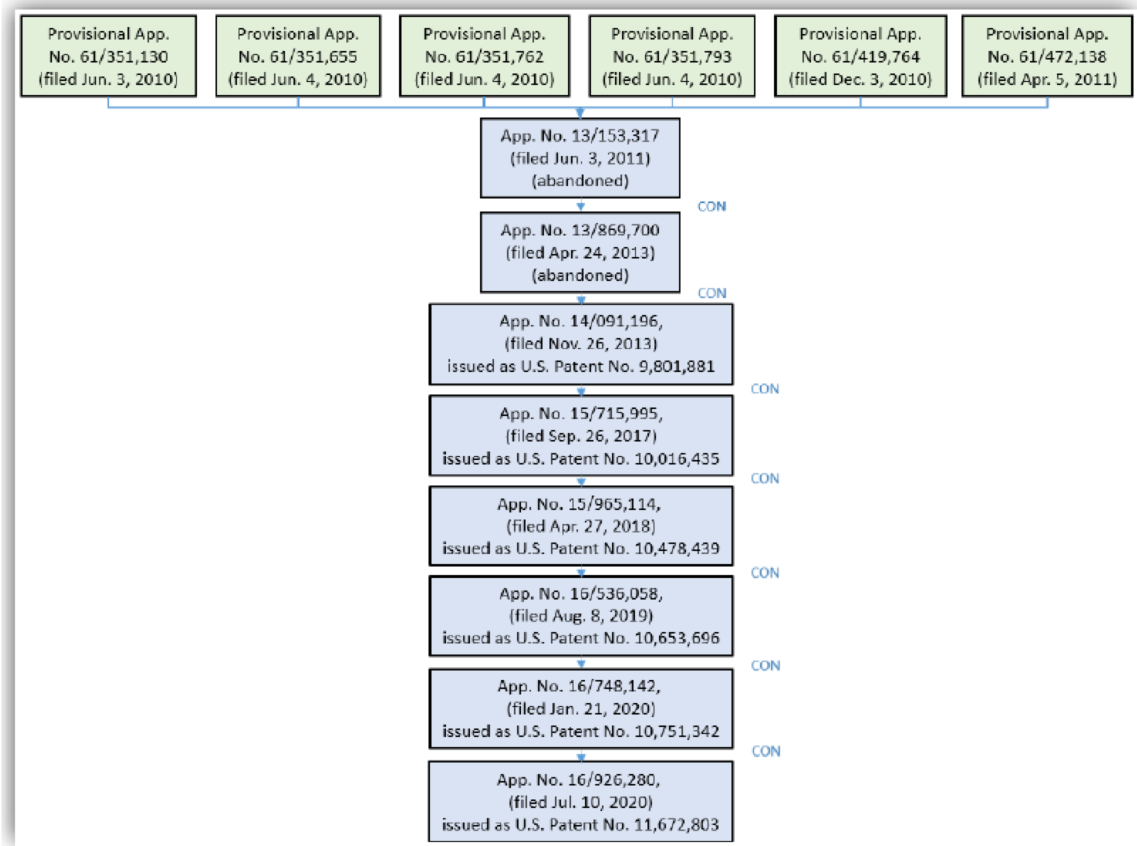
We determine no other claim terms need construction for purposes of this Decision. *See Wellman, Inc.*, 642 F.3d at 1361.

C. Effective Filing Date of the Challenged Claims

To be eligible for post-grant review, Petitioner has the burden to show that the effective filing date of at least one claim of the '803 Patent is on or after March 16, 2013. *See US Endodontics, LLC v. Gold Standard Instruments, LLC*, PGR2015-00019, Paper 17 at 9–12 (PTAB Jan. 29, 2016).

The '803 Patent issued from the '280 Application, filed on July 10, 2020. Ex. 1001, codes (21), (22). The patent family tree provided by Petitioner is reproduced below:

[10] Although Petitioner initially offered a different construction for "therapeutically effective amount" than the express definition in the Specification, Petitioner waived its argument during the oral hearing and accepted the Board's construction of the term set forth in the Institution Decision (Dec. Inst. 8–10), which is consistent with our construction here. Tr. 95:4–8.



Pet. 61. The family tree shows that the ’280 Application claims priority to seven non-provisional applications: U.S. Application Nos. 16/748,142; 16/536,058; 15/965,114; 15/715,995; 14/091,196; and 13/869,700, and the earliest-filed non-provisional application, the ’317 Application. *Id.*, code (63). The parties agree that the specifications of those non-provisional applications are substantively identical to the ’803 Patent Specification. Pet. 61; PO Resp. 3 n.1. As such, both Petitioner and Patent Owner cite to the Specification of the ’803 Patent in their written description discussions instead of the ’317 Application. Pet. 61 n.5; PO Resp. 3 n.1. For ease of reference, we do the same in this Decision with the understanding that all citations to the ’803 Patent Specification also apply to citations to the ancestral non-provisional application specifications.

Patent Owner asserts that the ’803 Patent claims are entitled to claim the benefit of the filing date of the ’317 Application, which was filed on June 3, 2011. PO Resp. 3–4 n.1; Ex. 1001, code (63). Petitioner disagrees, asserting that the Challenged Claims are not entitled to claim the benefit of any of the priority applications, therefore making the effective filing date of the Challenged Claims July 10, 2020, the filing date of the ’803 Patent itself. Pet. 3.

Having considered the arguments and evidence presented at trial, we determine Petitioner has shown by a preponderance of the evidence that the Challenged Claims are not entitled to claim priority to any earlier priority application because the claims lack written description support in the earlier applications. Thus, we determine that the effective filing date of the Challenged Claims is July 10, 2020, and the Challenged Claims are eligible for post-grant review. Our analysis follows.

1. Legal Standards

To claim the benefit of an earlier filing date of a U.S. patent application under 35 U.S.C. § 120, the claimed invention must be disclosed “in the manner provided by § 112(a) (other than the requirement to disclose the best mode)” in the earlier application. *See* 35 U.S.C. § 120. Thus, for the ’803 Patent to be eligible for post-grant review, Petitioner must show that at least one claim does not have adequate written description support or is not enabled in an ancestor application filed before March 16, 2013.

The test for written description support is “whether the disclosure of the application relied upon reasonably conveys to those skilled in the art that the inventor had possession of the claimed subject matter as of the filing date” based on an “objective inquiry into the four corners of the specification.” *Ariad Pharms., Inc. v. Eli Lilly & Co.*, 598 F.3d 1336, 1351

(Fed. Cir. 2010) (en banc). The written description requirement is satisfied when the specification "set[s] forth enough detail to allow a person of ordinary skill in the art to understand what is claimed and to recognize that the inventor invented what is claimed." *Univ. of Rochester v. G.D. Searle & Co.*, 358 F.3d 916, 928 (Fed. Cir. 2004).

The specification need not provide exact or verbatim textual support for the claimed subject matter at issue. *Fujikawa v. Wattanasin*, 93 F.3d 1559, 1570 (Fed. Cir. 1996). Moreover, "the written description requirement does not demand either examples or an actual reduction to practice." *Ariad Pharms.*, 598 F.3d at 1352. "[A]n applicant is not required to describe in the specification every conceivable and possible future embodiment of his invention." *Cordis Corp. v. Medtronic AVE, Inc.*, 339 F.3d 1352, 1365 (Fed. Cir. 2003). Furthermore, "[a] specification may . . . contain a written description of a broadly claimed invention without describing all species that [the] claim encompasses." *Id.*

That said, the Federal Circuit explains that written description of a genus requires more than a "broad outline of a genus's perimeter":

> [W]ritten description of a broad genus requires description of not only the outer limits of the genus but also of either a representative number of members of the genus or structural features common to the members of the genus, in either case with enough precision that a relevant artisan can visualize or recognize the members of the genus.

Regents of the Univ. of Minn. v. Gilead Sciences, Inc., 61 F.4th 1350, 1356 (Fed. Cir. 2023). "For genus claims, . . . we have looked for blaze marks within the disclosure that guide attention to the claimed species or subgenus." *Id.*

The written description inquiry is a question of fact, is context specific, and must be determined on a case-by-case basis. *Ariad Pharms.*, 598 F.3d at 1351 (citing *Ralston Purina Co. v. Far-Mar-Co, Inc.*, 772 F.2d 1570, 1575; *Capon v. Eshhar*, 418 F.3d 1349, 1357–1358 (Fed. Cir. 2005)). "[T]he level of detail required to satisfy the written description requirement varies depending on the nature and scope of the claims and on the complexity and predictability of the relevant technology." *Id.* (citing *Capon*, 418 F.3d at 1357–58). Factors used to evaluate the sufficiency of a disclosure include: 1) "the existing knowledge in the particular field"; 2) "the extent and content of the prior art"; 3) "the maturity of the science or technology"; and 4) "the predictability of the aspect at issue." *Id.* (citing *Capon*, 418 F.3d at 1359).

"Patent claims are awarded priority on a claim-by-claim basis based on the disclosure in the priority applications." *Lucent Techs., Inc., v. Gateway, Inc.*, 543 F.3d 710, 718 (Fed. Cir. 2008) (citations omitted). We address below whether the challenged claims have written description support in an ancestor application filed before March 16, 2013.

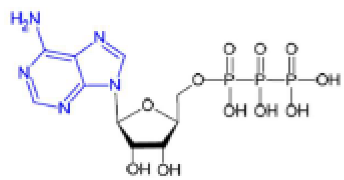
2. Technical Background

Independent claims 1 and 8 recite a method for treating CLL or SLL comprising administering a therapeutically effective amount of a BTK inhibitor with the requisite structure and function. Ex. 1001, claims 1 and 8. BTK is a protein kinase, which catalyzes the transfer of phosphate from an adenosine triphosphate ("ATP") molecule to other proteins to modulate their activity in the body. Ex. 2012 ¶ 21.

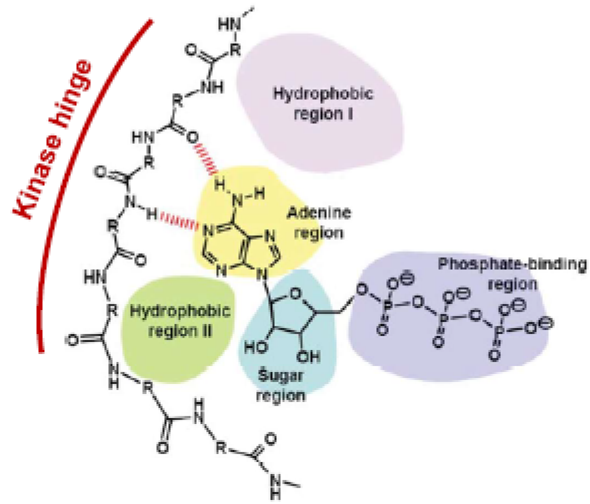
a) Active Site Binding in Protein Kinases

ATP binds the protein kinase at the "active site." *Id.* By 2011, approximately 1300 structures of protein kinases with bound ligands were publicly available in the Protein Data Bank ("PDB"), which is a computer archival service that was freely available to the public. Ex. 2010 ¶ 46 n.3. Ten of those recorded structures were of BTK with different ligands. *Id.* ¶ 46. A POSA in 2011 would have understood that the kinase active site is "highly conserved between all protein kinases." *Id.* (citing Ex. 2028,[11] 1).

Reproduced below, Patent Owner provides an annotated illustration of ATP and ATP binding to the active site of a kinase:



ATP



ATP binding to kinase (EX2048.0001) (annotated)

PO Resp. 15; Ex. 2012 ¶ 22. The structure of ATP is shown above on the left with the adenine portion of ATP in blue. The ATP binding site of kinase is shown above on the right with the kinase hinge region labeled in red between hydrophobic regions I and II shown in pink and green circles, respectively. The adenine region, sugar region, and phosphate-binding region of the kinase and the respective bound portions of ATP are shown in

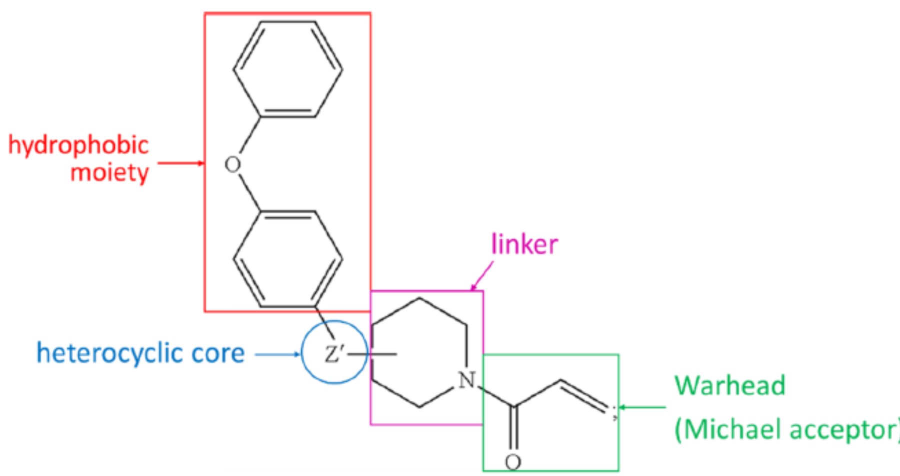
---

[11] Akritopoulou-Zanze et al., *Kinase-Targeted Libraries: The Design and Synthesis of Novel, Potent, and Selective Kinase Inhibitors*, 14 DRUG DISCOVERY TODAY 291–97 (2009). Ex. 2028.

yellow, blue, and purple circles, respectively. Finally, the hydrogen bonds between the adenine portion of ATP and the kinase hinge residues are shown as red dashed lines. Ex. 2010 ¶ 61; Ex. 2012 ¶ 22; Ex. 2048,[12] 1.

b) Protein Kinase Inhibitors

Numerous kinase inhibitors include scaffolds that mimic the bonding of ATP to the hinge residues. Ex. 2010 ¶¶ 63, 67, 83–89; Ex. 2012 ¶¶ 22, 27, 28. Kinase inhibitors typically form one to three hydrogen bonds with the kinase hinge residues. Ex. 2010 ¶ 63; Ex. 1052,[13] 20. Irreversible BTKIs generally comprise four components: (1) a hydrophobic moiety, (2) a heterocyclic core (or Z′-scaffold or Z′-structure) (optionally substituted), (3) a linker, and (4) a warhead, as shown by the following annotated structure of the challenged claims:



annotated structure of the Challenged Claims

---

[12] Le Corre et al., *Synthesis and Biological Evaluation of a Triazole-Based Library of Pyrido*[*2,3-d*]*Pyrimidines as FGFR3 Tyrosine Kinase Inhibitors*, 8 ORG. BIOMOL. CHEM. 2164–73 (2010). Ex. 2048.

[13] Ghose et al., *Knowledge Based Prediction of Ligand Binding Modes and Rational Inhibitor Design for Kinase Drug Discovery*, 51 J. MED. CHEM. 5149–71 (2008). Ex. 1052 ("Ghose").

Ex. 1003 ¶¶ 93, 94; Ex. 2010 ¶ 70. The annotated structure of the challenged claims shows a phenoxyphenyl hydrophobic moiety in a red box, a Z′-structure heterocyclic core in a blue circle, a piperidine linker in a purple box, and a warhead (i.e., a Michael acceptor, which forms the covalent bond to BTK) in a green box.

BTK has a unique cysteine near the ATP-binding site (i.e., the active site) called Cys481, which eleven of approximately 500 kinases have. Ex. 2010 ¶ 72 n.7 (citing Ex. 1060,[14] 1–2). The warhead is attached to the heterocyclic core via the linker, which positions the warhead to form a covalent bond with Cys481. Ex. 2010 ¶ 49.

The hydrophobic moiety binds to the hydrophobic regions of kinases (i.e., the "specificity pockets" or "back pockets"). *Id.* ¶¶ 58, 59. Because the size and accessibility of the back pockets differ between kinases, the choice of hydrophobic moiety of a BTKI affects its specificity and potency. *Id.* ¶¶ 55–59.

The Z′-structure engages the hinge region of BTK via hydrogen bonds. Ex. 1003 ¶ 60. It was understood that a BTKI first reversibly interacts with the active site of a kinase through hydrogen bonds with the hinge region, which determines the orientation of the inhibitor within the kinase. *Id.* (citing Ex. 1049,[15] 1).

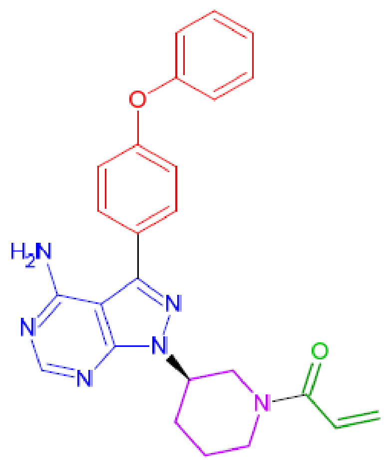
One example of an irreversible BTKI is ibrunitib, marketed as IMBRUVICA by Patent Owner. Ibrutinib had been disclosed in patent

---

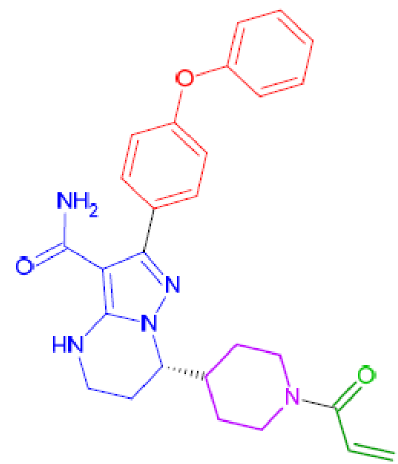
[14] Pan et al., *Discovery of Selective Irreversible Inhibitors for Bruton's Tyrosine Kinase*, 2 CHEMMEDCHEM 58–61 (2007). Ex. 1060 ("Pan").

[15] Garuti et al., *Irreversible Protein Kinase Inhibitors*, 18 CURRENT MEDICINAL CHEM. 2981–94 (2011). Ex. 1049.

applications filed before 2011. Ex. 2025.[16] Another example of an iBTKI is zanubrutinib, marketed as BRUKINSA by Petitioner. Both are FDA approved to treat CLL/SLL and have the following structures, annotated in the same color scheme as the annotated structure above:



IMBRUVICA
(ibrutinib)



BRUKINSA
(zanubrutinib)

Pet. 10 (citing Ex. 1003 ¶ 39); Ex. 1003 ¶¶ 34, 38; Ex. 2010 ¶ 70. The annotated structures show ibrutinib and zanubrutinib have the same phenoxyphenyl hydrophobic moiety in red and the same warhead in green, but different Z′-structures in blue and differently oriented piperidine linkers in purple.

There are a number of structural differences between ibrutinib and zanubrutinib, including: (1) the aromaticity of the fused heterocyclic ring system (i.e., ibrutinib's ring system is biaryl and zanubrutinib's is non-biaryl[17]) (Ex. 1003 ¶¶ 40–41); (2) the heteroatom pattern of the ring system

[16] Honigberg et al., US 2008/0108636 A1, published May 8. 2008. Ex. 2025.

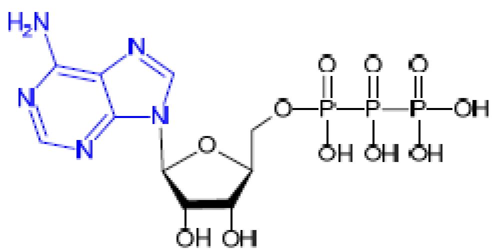
[17] The parties disagree on the '803 Patent's definition of "biaryl" and whether it refers to fully aromatic structures, as Petitioner asserts, or

(i.e., ibrutinib has four N heteroatoms at the C1, C2, C5, and C7 positions and zanubrutinib has three N heteroatoms at the C1, C4, and C8 positions) (*id.* ¶ 42); (3) the substitution pattern on the ring system (i.e., ibrutinib has an amino substitution at C4 of its 6-membered ring, whereas zanubrutinib has an amide substitution at the C3 position of its 5-membered ring) (*id.* ¶¶ 43–45); and (4) the linker (i.e., in ibrutinib the linker is connected to the N1 atom of the ring system at C3 of the piperidine group, and in zanubrutinib the linker is connected to C7 of the ring system via C4 of the piperidine group) (*id.* ¶¶ 46–47).

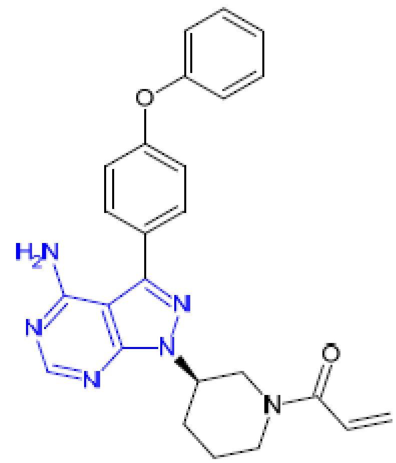
Ibrutinib is designed to mimic adenine (the ring system of adenosine triphosphate (ATP)) so that ibrutinib can bind to BTK in a mode analogous to adenine. Ex. 1003 ¶¶ 49–55. The parties dispute whether zanubrutinib mimics adenine, as its ring system is non-aromatic and has no amino substitution. *See* Pet. 14 (asserting zanubritinib does not mimic adenine) (citing Ex. 1003 ¶¶ 49, 52); PO Resp. 2 (stating zanubritinib mimics adenine). Notwithstanding which party is correct, the annotated structures comparing ATP to IMBRUVICA (ibrutinib) and BRUKINSA (zanubrutinib) are reproduced below:

---

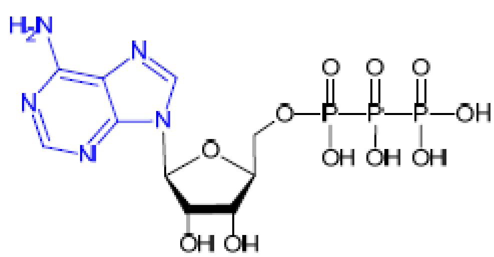
encompasses partially aromatic structures, as Patent Owner asserts. Pet. 67 n.8; PO Resp. 16. The parties' experts agree, however, that the literal definition of "biaryl" is two fully aromatic rings. Ex. 1003 ¶ 41; Ex. 1110, 82:2–6. We apply that definition here when we use the term "biaryl," but we note that our result would be the same even if we applied Patent Owner's definition, as the Specification fails to describe the full scope of the claims whether "biaryl" is defined as fully aromatic or not.



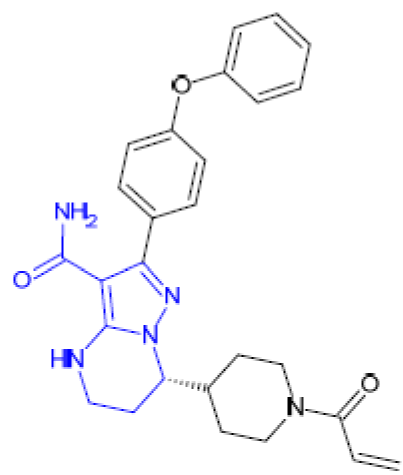
ATP



IMBRUVICA



ATP



BRUKINSA

The annotated structures depict the adenine of ATP in blue and the Z′-structures of ibrutinib and zanubrutinib in blue. The adenine of ATP and the Z′-structure of ibrutinib are both aromatic with an amino substitution at the C4 position of the 6-membered ring. Ex. 1003 ¶¶ 54, 55. The Z′-structure of zanubrutinib, on the other hand, has an amide substitution on the 5-membered ring and the phenoxyphenyl hydrophobic moiety is projected from a different position on the 5-membered ring with the linker located on the non-aromatic 6-membered ring. *Id.* ¶¶ 49, 52.

Ibrutinib and zanubrutinib have different hydrogen bond donors and acceptors on the Z′-structure and engage in different hydrogen bonds with BTK. *Id.* ¶ 61. A chart summarizing the hydrogen bond interactions of the two BTKIs is reproduced below:

IMBRUVICA-Btk complex		BRUKINSA-Btk complex	
Btk residue	IMBRUVICA moiety	Btk residue	BRUKINSA moiety
Thr 474	amino substituent (H bond donor)	Glu 475	$-NH_2$ of amide substituent (H bond donor)
Glu 475	amino substituent (H bond donor)	Met 477	$-C{=}O$ of amide substituent (H bond acceptor)
Met 477	N5 heteroatom (H bond acceptor)	Met 477	N4 heteroatom (H bond acceptor)

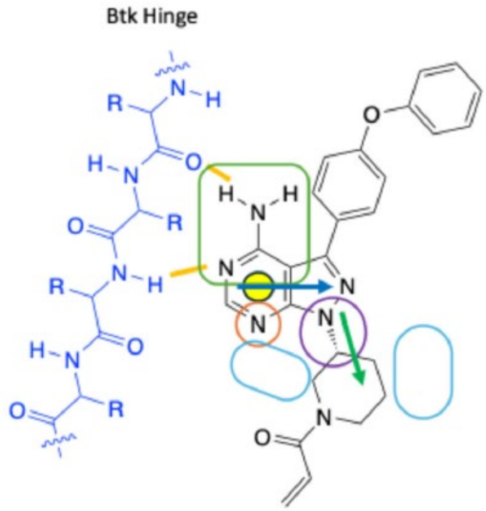
Id. ¶¶ 61–67 (summarizing X-ray crystallography data from Ex. 1053,[18] 5; 1076,[19] 6).[20]

Reproduced below is a figure provided by Dr. Veal of the binding mode and interactions of ibrutinib and zanubrutinib:

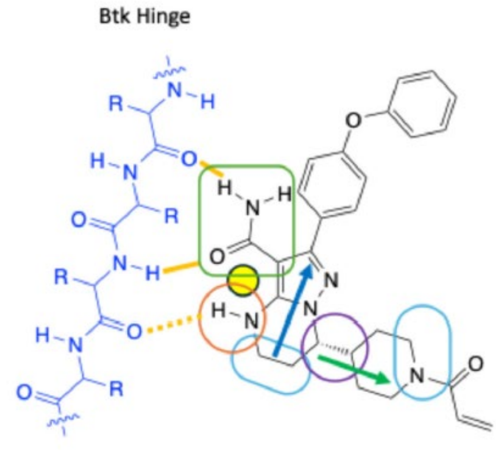
[18] Guo et al., *Discovery of Zanubrutinib (BGB-3111), a Novel, Potent, and Selective Covalent Inhibitor of Bruton's Tyrosine Kinase*, 62 J. MED. CHEM. 7923–40 (2019). Ex. 1053 ("Guo").

[19] Zhang et al., *Recent Advances in BTK Inhibitors for the Treatment of Inflammatory and Autoimmune Diseases*, 26 Molecules 4907 (2021). Ex. 1076 ("Zhang").

[20] Of note, a different group studying the same crystallography data of ibrutinib and zanubrutinib reported different hydrogen bonding patterns than Guo and Zhang, but the patterns still differ. *Id.* ¶ 63 (citing Ex. 1044,[20] 3–5).



Imbruvica

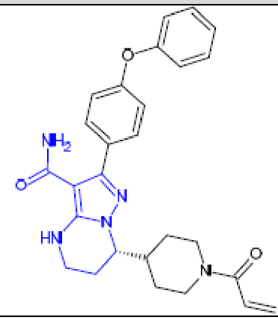
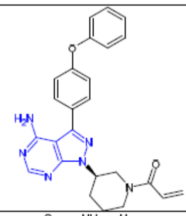
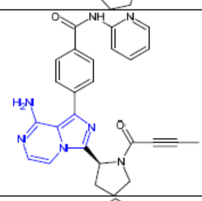
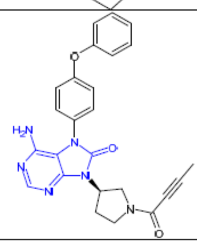
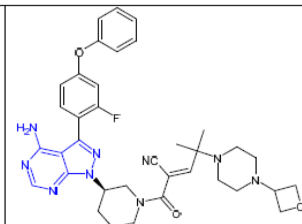
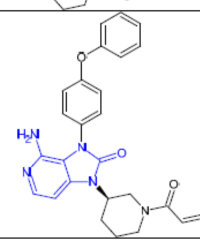


Brukinsa

Ex. 1106 ¶ 74. The figure illustrates as orange lines the two hydrogen bonds of ibrutinib on the left and the three hydrogen bonds of zanubrutinib on the right.

As of 2011, a POSA would have been aware of other compounds that inhibit BTK. *See* Ex. 1073, 8–9, Table 2. Moreover, Marcotte[21] disclosed the crystal structure of BTK reversibly bound to a compound (B43), which is similar to ibrutinib but without the warhead. In a 2021 survey of reported irreversible BTKIs, six BTKIs had a 5:6 fused heterocyclic ring system, like the challenged claims. Ex. 1003 ¶¶ 49–52; Ex. 1071, 6. The annotated structures of those six BTKIs are reproduced below:

[21] Marcotte et al., *Structures of Human Bruton's Tyrosine Kinase in Active and Inactive Conformations Suggest a Mechanism of Activation for TEC Family Kinases*, 19 PROTEIN SCIENCE 429–39 (2010). Ex. 2001 ("Marcotte").

Name	Date of First Approval	Structure
Zanubrutinib (BRUKINSA)	November 2019 (FDA)	
Ibrutinib (IMBRUVICA)	November 2013 (FDA)	
Acalabrutinib (CALQUENCE)	October 2017 (FDA)	
Tirabrutinib	March 2020 (Japan)	
Rilzabrutinib	Pending	
Tolebrutinib	Pending for multiple sclerosis	

Ex. 1003 ¶ 49. The annotated structures of the BTKIs show the Z′-scaffolds in blue. Each of the BTKIs except zanubrutinib has a 5:6 heterocyclic ring system with an amino substitution on C4 of the 6-membered ring adjacent to a nitrogen heteroatom (corresponding to N5 of ibrutinib), a hydrophobic moiety on the 5-membered ring (corresponding to C3 of ibrutinib), and a linker-warhead substitution on the 5-membered ring (corresponding to N1 of ibrutinib).

3. *Written Description Analysis of Claims 1 and 8*

Independent claims 1 and 8 recite a method for treating CLL or SLL comprising administering a therapeutically effective amount of an iBTKI with certain requisite "Structural Limitations" and "Functional Limitations." The Structural Limitations of the claims recite a phenoxyphenyl hydrophobic moiety, a piperidine linker, a warhead and a generic Z′-structure consisting of a 5:6 heterocyclic ring comprising 2–4 nitrogen heteroatoms with at least one nitrogen per ring. Ex. 1001, claims 1 and 8. The Functional Limitations require administering a "therapeutically effective amount" of the iBTKI, resulting in lymphocytosis, and forming a covalent bond with a residue of the iBTKI, such as Cys481. *Id.* Thus, the claims recite a genus of iBTKIs that must satisfy both the Structural and Functional Limitations of the claims.

To satisfy written description for the claimed genus of iBTKIs, the Specification must describe either a representative number of members of the genus or structural features common to the members of the genus with enough precision that a POSA can visualize or recognize the members of the genus. *See Ariad*, 598 F.3d at 1350. Patent Owner concedes that the Specification does not describe a representative number of members of the genus, stating "this is a common structural features case." Tr. 65:25–66:3. We, therefore, focus our analysis on whether the Specification adequately describes common structural features of the genus. We find that it does not.

We agree with Petitioner that the claims encompass a very large number of possible species in light of the various optional substitutions of the Z′-structure. As explained by Dr. Showalter, there are at least hundreds of possible ring systems, and each of those hundreds of ring systems can have optional substitutions selected from 21 broad categories, and each of those categories include numerous possible substitutions. Ex. 1003 ¶¶ 148–174. Accordingly, Dr. Showalter states that the "number of structural species encompassed by these Structural Limitations is at least in the billions." *Id.* ¶ 142. Patent Owner argues that Dr. Showalter ignores the full context of the claimed iBTKIs, including the fixed linker-warhead and phenoxyphenyl substituents on the Z′-structure. Patent Owner further asserts that those required substituents, their orientation, and the steric limitations that allow for binding to the BTK hinge would limit the scope of the Z′-scaffolds. PO Resp. 54–55 (citing Ex. 2010 ¶¶ 180–182). In its Reply, Petitioner narrows the potential structures that encompass the Z′-structure to address Patent Owner's criticisms of Dr. Showalter's calculations, but notes that the claims still cover over four million potential iBTKIs. Pet. Reply 5; Ex. 1105 ¶¶ 2, 31–37; Ex. 1106 ¶¶ 3, 185, 247, 251, 253–258, 264.

Whatever the precise number of potential Z′-structures may be, that number is very large. But the number of possible structures is not the only factor to consider. As Patent Owner notes, the claims must be considered as a whole. PO Resp. 40 (citing *Novozymes A/S v. DuPont Nutrition Biosciences APS*, 723 F.3d 1336, 1349 (Fed. Cir. 2013)). Specifically, the claims also require that the BTKI satisfy the Functional Limitations, including the covalent bonding of Cys481. Thus, the Specification must describe the structural features common to members of the genus that satisfy both the Structural and Functional Limitations of the claims. *Ariad*, 598 F.3d at 1350.

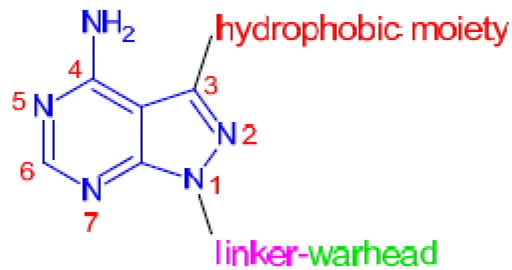
For generic claims, the Federal Circuit has set forth several factors for evaluating the adequacy of the disclosure, including "the existing knowledge in the particular field, the extent and content of the prior art, the maturity of the science or technology, [and] the predictability of the aspect at issue." *Ariad*, 598 F.3d at 1351 (quoting *Capon v. Eshhar*, 418 F.3d 1349, 1359 (Fed. Cir. 2005)). Petitioner asserts that the Z′-structure "unpredictably affects a BTKI's bioactive orientation, and thereby unpredictably impacts whether the BTKI will covalently bind Btk and inhibit Btk irreversibly." Pet. 16 (citing Ex. 1003 ¶¶ 53–76). Moreover, Petitioner asserts that the structure-activity relationship ("SAR") between the Z′-structure and the function of covalently binding Btk is not fully understood, even today, especially for compounds that do not include close structural similarity to adenine. *Id.*; Ex. 1003 ¶ 54.

Patent Owner and its expert Dr. Eck assert the opposite. According to Patent Owner, the field of kinase inhibitors was mature and well-developed. PO Resp. 12; Ex. 2010 ¶ 42; Ex. 2010 ¶ 46. Moreover, Dr. Eck opines that a POSA would have understood that Z′-scaffolds bind within BTK's highly conserved hinge region while orienting the fixed linker-warhead and phenoxyphenyl moiety and are "fungible and predictable." Ex. 2010 ¶¶ 83–84. Moreover, a POSA would have known that binding was predictable, particularly in light of the "advanced understanding of the structure of the kinase active site" coupled with the data from Marcotte and computer modeling software available in 2011. *See id.* ¶¶ 46, 116.

Having considered the parties' respective arguments and evidence, we agree with Patent Owner that the field of kinase inhibitors was not new and that the kinase active site was known to be highly conserved. *See* Ex. 2010 ¶ 45 (citing Ex. 2028, 1; Ex. 1075, 3). That said, we are not persuaded that the existing knowledge in 2011 rose to such a level of predictability that would require only minimal disclosure in the Specification for purposes of written description. *Ariad*, 598 F.3d at 1351. Patent Owner and its expert Dr. Eck rely heavily on Marcotte to assert that hinge-binding across all "ibrutinib-like BTKIs" was predictable. PO Resp. 21–22; Ex. 2010 ¶¶ 111–112. But Patent Owner broadly defines "ibrutinib-like" to include not only compounds with the same Z′-scaffold as ibrutinib, but any Z′-scaffold that falls within the scope of the claims. PO Resp. 4 n.2 (using "ibrutinib-like iBTKIs" to refer to iBTKIs "having the structural features (based on ibrutinib) recited in the claims: fixed phenoxyphenyl and linker-warhead substituents attached to a central 5:6 heterocyclic scaffold with nitrogen heteroatoms"). We agree with Petitioner that that definition is too broad, particularly given that Marcotte only provides data on the Z′-scaffold of ibrutinib. Ex. 2001, 3, Fig. 2(A) (providing chemical structure of B43).

Turning to the disclosure, the Specification does not expressly disclose the generic Markush formula of the Challenged Claims or the recited Z′-structure. *See generally* Ex. 1001; *see also* Ex. 1003 ¶ 97. Indeed, we agree with Petitioner that the only compounds expressly disclosed by the Specification that satisfy the Structural Limitations are ibrutinib and its positional/chiral isomers. Ex. 1001, 65:10–25, 71:1–20, 77:10–25, 79:50–65, 80:1–15, 80:37–57 (compounds 4, 9, 13, 14); Ex. 1003 ¶¶ 102, 188. The remaining compounds disclosed by the Specification do not include the hydrophobic moiety or linker-warhead required by the claims. Ex. 1003 ¶¶ 102, 188.

Formulas (A)–(D) and (F) cover compounds with the Z′-structure of ibrutinib or example E6 of Formula (E). Ex. 1001, 47:24–35 (Formula (A)), 51:25–40 (Formula (B)), 56:10–25 (Formula (C)), 58:50–65 (Formula (D)), 63:25–40 (Formula (F)); Ex. 1003 ¶¶ 103–111, 116, 188. As explained by Dr. Showalter, Formulas (B)–(D), (F), and ibrutinib have the same structure:



Formula (B)-(D), (F)
and IMBRUVICA

Ex. 1003 ¶ 110. As seen by the structure, each Formula has a linker-warhead at the N1 position, a hydrophobic moiety at the C3 position of the 5-membered ring, and an amine substitution at the C4 position of the 6-membered ring, like ibrutinib. *Id.*

Regarding Formula (E), the Specification simply states that the Z′-structure (i.e., the "orb") "is a moiety that binds to the active site of a kinase, including a tyrosine kinase, further including a Btk kinase cysteine homolog." Ex. 1001, 62:15–17. The Specification does not explain what structural features are necessary to satisfy the functional requirement of binding to the active site of a kinase. At best, the Specification provides eight embodiments for the orb of Formula (E), i.e., examples E1–E8. *Id.* at 62:44–63:20. But the Specification does not explain which of those embodiments—besides ibrutinib's E7 scaffold—would result in irreversibly binding the Cys481 residue of the hinge region when paired with the fixed structural features of the claims. Moreover, as noted by Petitioner, the Specification "does not describe any BTKI resembling [zanubrutinib]." Pet. 65 (citing Ex. 1003 ¶¶ 194–208).

In light of this limited disclosure, we agree with Dr. Showalter that the Specification fails to sufficiently describe the structural features common to all members of the genus sufficient for a POSA to visualize the full scope of the challenged claims. *See* Ex. 1003 ¶¶ 39–48, 98–117, 188–207. Specifically, the Specification does not describe common structural features of the BTKIs that satisfy the Structural Limitations and the functional requirement to covalently bind Cys481 or another residue of BTK. *Id.* ¶¶ 211–219. Moreover, there are insufficient blaze marks in the Specification that would guide a POSA from the generic structures disclosed in the Specification to the full scope of Z′-structures that also satisfy the Functional Limitations. *Id.* ¶¶ 112–114, 208.

We agree with Petitioner that this case is analogous to *Idenix Pharmaceuticals LLC v. Gilead Sciences Inc.*, 941 F.3d 1149 (Fed. Cir. 2019). There, the involved patent claims a method of treating hepatitis C virus ("HCV") with an effective amount of a nucleoside with a specific chemical and stereochemical structure. *Id.* at 1154. Specifically, the claims require a methyl at the 2'-up position, but allow for nearly any substituent at the 2'-down position of the nucleoside. *Id.* at 1155. Accordingly, it was undisputed that there are at least many thousands of potential nucleosides that meet the structural limitations of the claim. *Id.* at 1157. Moreover, because the claims require administering an "effective amount," the claims further require the use of some set of compounds that are effective for treating HCV. *Id.* at 1155. The patent discloses eighteen formulas that are "principal embodiments" of the compounds that may treat HCV, but the Federal Circuit found that the specification "provides no indication that any nucleosides outside of those disclosed in its formulas could be effective to treat HCV—much less any indication as to *which* of those undisclosed nucleosides would be effective." *Id*. at 1164. Moreover, the court found that the absence of the accused product was "conspicuous" in light of the thousands of possible nucleosides listed substituent-by-substituent in the various positions. *Id.* at 1165. Thus, the Federal Circuit held the patent lacked adequate written description support because it failed to provide "sufficient blaze marks to direct a POSA to the specific subset of 2'-methyl-up nucleosides that are effective in treating HCV." *Id*. at 1164.

Similarly, here, the Specification discloses thousands of potential Z′-structures, but does not explain "what makes them effective, or why." *Id.* Without that guidance, "the listed examples and formulas cannot provide adequate written description support for undisclosed [compounds that satisfy the Functional Limitations of the claims.]" *Id.* Moreover, it is undisputed that zanubrutinib is not described in the Specification, as zanubrutinib, like the accused product in *Idenix*, was disclosed after the filing date of the patent. *Id.*; Pet. 1 (stating zanubrutinib was disclosed in 2015). Zanubrutinib's Z′-structure, however, is not taught by any of the potential Z′-structures described in the Specification and is, likewise, "conspicuously absent." *Id.* at 1165.

We have considered Patent Owner's arguments and evidence, but do not find them persuasive. For example, Patent Owner argues that the Specification describes common structural features for the iBTKIs used in the claimed method. PO Resp. 39. Patent Owner asserts that the genus of claimed compounds are "ibrutinib-like iBTKIs with two ***fixed*** components from the specification's preferred embodiment, ibrutinib—a linker-warhead and a phenoxyphenyl moiety—and one ***structurally-constrained*** component—the hinge-binding Z′-scaffold." *Id.* According to Patent Owner, the Specification discloses the fixed linker-warhead as binding with BTK's Cys481 and driving irreversible inhibition. *Id.* at 40–42 (citing Ex. 1001, 45:56–60, 46:10–12; Ex. 2010 ¶¶ 131–135). Patent Owner asserts that a POSA would also understand that the phenoxyphenyl moiety binds within the “specificity pocket” and conveys inhibitor selectivity and potency. *Id.* at 42–43 (citing Ex. 1001, 6:9–80:35; Ex. 2010 ¶¶ 142–144).

We note, however, that the Specification must describe structural features common to the members of the *claimed genus* so that a POSA can visualize or recognize the members of the genus. *Ariad*, 598 F.3d at 1350. To the extent Patent Owner relies on the two fixed structural limitations of the claims to establish common structural features for written description of the genus, we are not persuaded. It is the variability of the recited Z′-structure that defines the genus of claimed BTKIs. In other words, although the structure of the Z′-scaffold may be “structurally-constrained” by the fixed linker-warhead and phenoxyphenyl moiety (PO Resp. 39), the scope of the genus of BTKIs is defined by the Z′-structure and its optionally substituted 5:6 heterocyclic ring system with 2–4 nitrogen heteroatoms. Ex. 1001, claims 1 and 8. Thus, the common structural features of the claimed genus for purposes of written description support must necessarily relate to the structure of the variable Z′-scaffold.

Patent Owner, however, has only defined the claimed genus by the functional features of the Z′-scaffold. In particular, Patent Owner asserts that the Z′-scaffold sets the orientation of the fixed components and forms hydrogen bonds to the highly conserved kinase hinge, which both “guide a POSA to choose appropriate scaffolds for compounds to use in the claimed methods.” PO Sur-reply 5; PO Resp. 21–24, 43–50. Such functional definitions of the genus, however, are not sufficient. “[T]he specification must demonstrate that the applicant has made a generic invention that achieves the claimed result and do so by showing that the applicant has invented species sufficient to support a claim to the functionally-defined genus.” *Ariad*, 598 F.3d at 1349; *see also Eli Lilly*, 119 F.3d at 1568 (finding inadequate written description where specification did not distinguish the genus from other materials in any way except by function and thus provided "only a definition of a useful result rather than a definition of what achieves that result").

Patent Owner asserts that a POSA would understand that the genus of iBTKIs would be discernible because a POSA would be able to identify which of the possible Z′-structures would covalently bond Cys481 by using available modeling software. PO Resp. 42 (citing Ex. 2001, 3; Ex. 2010 ¶ 139). Specifically, Dr. Eck opines that a POSA would have understood that Z′-scaffolds that bind to the highly conserved hinge region are "fungible and predictable," and that the scaffolds capable of making those hinge interactions are "diverse." Ex. 2010 ¶ 84; *see also id.* ¶¶ 85–89. Patent Owner argues that the Specification provides example scaffolds of substituted 5:6 fused heterocyclic scaffolds (i.e., E1–E8) and that a POSA would have been aware of many scaffolds that bind to the hinge region. PO Resp. 44–45 (citing Ex. 1001, 62:44–63:21; Ex. 2010 ¶¶ 149–152).

Moreover, according to Patent Owner, a POSA would have known the precise binding mode of ibrutinib-like iBTKIs based on Marcotte. *Id.* at 46 (citing Ex. 2010 ¶¶ 70, 71, 111, 112). Patent Owner asserts that Marcotte "provides detailed information about the geometry and specific amino acid interactions between a compound with ibrutinib-like binding and BTK." *Id.* (citing Ex. 2001; Ex. 2010 ¶¶ 70, 71, 105, 111, 112). Patent Owner argues that this information shows how (1) the linker-warhead can project towards BTK's Cys481; (2) the phenoxyphenyl orients and binds within BTK's back pocket; and (3) the Z′-scaffold hydrogen bonds to the hinge residues. *Id.* at 46–47 (citing Ex. 2001; Ex. 2010 ¶¶ 71, 111, 112). According to Patent Owner, a POSA in 2011 would have used software with Marcotte's data to visualize the structures within the claims to recognize the compounds that bind and inhibit BTK. *Id.* at 47 (citing Ex. 2010 ¶¶ 111–118, 153).

We are not persuaded. As explained above, Marcotte is limited to the ibrutinib Z′-scaffold and does not provide binding data for any other scaffolds (that also include the fixed structural components), which would be encompassed by the challenged claims. Ex. 2001; Ex. 1106 ¶ 55. Moreover, the Specification is silent on what structural features are common to the genus of Z′-scaffolds that enable hydrogen bonding and covalent binding to Cys481. As Patent Owner admits, the fixed linker-warhead and phenoxyphenyl groups could be attached to "numerous diverse Z′-scaffolds with H-bond acceptors and donors arranged in the proper geometry to form the appropriate H-bonds with the hinge." PO Resp. 54; Ex. 2010 ¶¶ 145–153. But, as Petitioner notes, the Specification never mentions H-bonding in the context of the Z′-structure and does not distinguish the claimed compounds that covalently bind Btk from ones that do not. Pet. Reply 15–16; Ex. 1106 ¶¶ 50, 178, 182, 183, 212, 228. Moreover, we note the testimony of both parties' experts that even if a BTKI had each of the fixed structural features, a compound may not irreversibly inhibit Btk because other characteristics of the compound, "such as electrostatics, complementarity, and Van der Waals interactions, would impact the ability to covalently bind to Btk." Ex. 1106 ¶ 44 (Veal); Ex. 1110 (Eck), 160:2–16. Indeed, Dr. Veal notes that Compound 7 of Zhang, analyzed by Dr. Eck, satisfies the Structural Limitations of the claims, but experimental results show that Compound 7 is inactive. Ex. 1106 ¶ 111; Ex. 2063, 2 (noting Compound 7 has "weaker potency against BTK"). Thus, although compounds may satisfy the Structural Limitations of the claims but not the Functional Limitations, the Specification provides no guidance as to which

Z′-scaffolds would be effective, or why. *See Idenix*, 941 F.3d at 1164 (stating lists or examples of supposedly effective molecules are insufficient without explanation of "what makes them effective, or why").

In response, Patent Owner asserts that "'written description does not require a ***perfect*** correspondence' but a 'more modest[]' '***correlation*** between structure and function.'" PO Sur-reply 14 (quoting *Ajinomoto Co. v. Int'l Trade Comm'n*, 932 F.3d 1342, 1360 (Fed. Cir. 2019)). *Ajinomoto*, however, is distinguishable because there, the patent owner actually identified a correlation between structural features and function. Specifically, the evidence showed that deviations in the promoter toward the consensus sequence generally increased promoter strength. *Ajinomoto*, 932 F.3d at 1360. Here, Patent Owner has not provided any common structural features of the Z′-scaffold that would result in covalent binding of Cys481.

Patent Owner argues that modeling would identify those compounds not expected to inhibit BTK. PO Sur-reply 14. But the Specification is silent on the use of molecular modeling to determine which Z′-structures will bind the hinge region of BTK to result in the covalent binding of Cys481. Ex. 1106 ¶ 32. Moreover, as Petitioner's experts Dr. Veal and Dr. Showalter note, a POSA would not have relied on modeling to initially screen a library of compounds to identify which ones bind the target. *Id.* ¶¶ 33–34; Ex. 1105 ¶¶ 9–12, 14. This is consistent with Patent Owner's expert Dr. Scott's experience. Ex. 1117, 34:14–35:5 (testifying that he did not work with a modeler "as part of the initial screening effort" because "one of the values of modeling is when you have identified a specific compound of interest"). The experts also agree that running assays was necessary to determine whether a potential inhibitor would actually covalently bind Cys481. Ex. 1106 ¶ 36; Ex. 1117, 41:9–13 (Dr. Scott testifying that assays are "valuable to validate predictions of modeling with data"). Thus, we are not persuaded that a POSA in 2011 could have relied on modeling alone to identify the genus of iBTKIs that covalently bind Cys481.

Patent Owner also argues that a POSA could predictably use "scaffold swapping," where scaffolds known to bind to a kinase hinge are swapped between inhibitor structures. PO Resp. 23 (citing Ex. 2010 ¶¶ 103–107; Ex. 1077,[22] 7). But "[a] 'mere wish or plan' for obtaining the claimed invention is not adequate written description." *Centocor Ortho Biotech v. Abbott Lab'ys*, 636 F.3d 1341, 1348 (Fed. Cir. 2011) (quoting *Regents of the Univ. of Cal. v. Eli Lilly & Co.*, 119 F.3d 1559, 1566 (Fed. Cir. 1997)).

We find the Federal Circuit's opinion in *PureCircle USA Inc. v. SweeGen, Inc.*, No. 22-1946, 2024 WL 20567 (Fed. Cir. Jan. 2, 2024) to be instructive. There, the claims recite a method of producing a steviol glycoside called Reb X using UDPglucosyltransferase enzymes ("UGTs"). *Id.* at *1. The parties stipulated that the term "UGT" is defined by its function of transferring a glucose unit to a steviol glycoside molecule. As such, the claims recite a genus of UGT enzymes defined by its function. *Id.* at *2. The specification identifies only one UGT that could produce Reb X, but the genus of UGTs is very large. *Id.* The patent owner asserted that there were only five known enzymes capable of producing Reb X and that any mutations capable of the required synthesis could be determined with reasonable clarity through "homology modeling." *Id.* at *3. The patent owner also asserted that although the mutants would have to be tested to see

[22] Zhang et al., *Targeting cancer with small molecule kinase inhibitors*, 9 CANCER 28–39 (2009). Ex. 1077.

if they could produce Reb X, the patent owner claimed that such testing was routine. *Id.*

The Federal Circuit held that the specification did not sufficiently disclose common structural features of the members of the genus to satisfy the written description requirement. Like here, neither the specification nor the claims mention modeling for determining common structure. *Id.* And, like modeling of a BTKI, the Federal Circuit noted that homology modeling cannot be used to determine what structural features are common to enzymes capable of producing Reb X. *Id.* at *4. Homology modeling can only be used to determine common structures of known enzymes, not create structures of other unknown enzymes. *Id.* Moreover, as is the case here, trial and error testing after modeling would be required to identify potential active candidates. *Id.*; Ex. 1106 ¶ 36; Ex. 1117, 41:9–13. Thus, like the claims in *PureCircle*, we find the Challenged Claims "merely recite a description of the problem to be solved while claiming all solutions to it and . . . cover any compound later actually invented and determined to fall within the claim's functional boundaries—leaving it to the pharmaceutical industry to complete an unfinished invention." *Id.* at *4 (quoting *Ariad*, 598 F.3d at 1353).

Patent Owner relies heavily on the *UroPep* case to support its argument. *Erfindergemeinschaft UroPep GbR v. Eli Lilly & Co.*, 276 F. Supp. 3d 629 (E.D. Tex. 2017), *aff'd* 739 F. App'x 643 (Fed. Cir. 2018) ("*UroPep*"). There, the claims recite a method of administering an effective amount of an inhibitor of the enzyme phosphodiesterase V ("PDE") to treat benign prostatic hyperplasia. 276 F. Supp. 3d at 638. The parties agree that the claimed genus of PDE5 inhibitors is very large and that hundreds of selective PDE5 inhibitors were known in the art at the time of the invention.

Id. at 646. The disclosure does not expressly discuss the common structural features of PDE5 inhibitors, but UroPep presented undisputed evidence that a POSA would recognize the common structural features of the PDE5 inhibitor genus. *Id.* at 652–53. According to Patent Owner, like the state of the art for PDE5 inhibitors, "in 2011, a POSA would have had a mature, sophisticated understanding of the structure-function relationship for ibrutinib-like iBTKI's based on Marcotte." PO Resp. 63.

UroPep, however, is distinguishable from the facts of this case. Unlike the genus of PDE5 inhibitors, here, there were not hundreds of known iBTKIs that satisfy both the Structural and Functional Limitations of the claim. *See UroPep*, 276 F. Supp. 3d at 646. Indeed, by 2021, there were only six known iBTKIs with a 5:6 heterocyclic ring system. Ex. 1003 ¶¶ 49–52. Moreover, in *UroPep*, there was no evidence that a compound that selectively inhibits PDE5 would not effectively treat BPH and satisfy the functional limitation of the claim. 276 F. Supp. 3d at 650. Here, the parties' experts agree that compounds that satisfy the Structural Limitations will not necessarily satisfy the Functional Limitations of the claims. *See* Ex. 1106 ¶ 44 (Veal); Ex. 1110 (Eck), 160:2–16; *see also* Ex. 1106 ¶ 111; Ex. 2063, 2 (noting Compound 7 has "weaker potency against BTK"). Thus, we find the *Idenix* and *PureCircle* cases more analogous to the facts of this case than *UroPep*.

Patent Owner tries to distinguish the *Idenix* case by stating the claims were not limited by any particular enzyme function and, therefore, lacked explanation for what makes them effective. PO Resp. 63–64 (citing *Idenix*, 941 F.3d at 1164). But the claims in *Idenix* were limited to enzymes that were therapeutically effective for treating HCV. *Idenix*, 941 F.3d at 1155.

Thus, like the Specification here, the specification in *Idenix* failed to explain "what makes [the supposedly effective compounds] effective, or why." *Id.*

Patent Owner also attempts to distinguish *Idenix* and the fact that the accused product was "conspicuously absent" from the disclosure in the specification. PO Resp. 65–66 (quoting *Idenix*, 941 F.3d at 1165). Here, Patent Owner argues that zanubrutinib "aligns exactly with the specification's disclosure of structures that irreversibly inhibit BTK to treat CLL/SLL in patients and induce lymphocytosis." *Id.* at 66. Patent Owner's argument strains credulity. Zanubrutinib—including its Z′-scaffold—is clearly not disclosed by the Specification, despite the various Formulas (A)–(F) and Z′-scaffold examples E1–E8 taught by the Specification. *See* Ex. 1001, 47:5–80:36. Moreover, it is telling that Formulas (B)–(D) and (F) of the Specification and each of the approved or preclinical BTKIs under investigation other than zanubrutinib have the same or similar Z′-structure as ibrutinib, which mirrors the structure of ATP. Ex. 1003 ¶¶ 49, 110. We therefore credit the testimony of Dr. Showalter that zanubrutinib, which was disclosed for the first time in 2015, has a unique Z′-structure that is not described by the Specification and, therefore, is "conspicuously absent." *Id.* ¶¶ 49, 102; *see also Idenix*, 941 F.3d at 1165.

Having considered the evidence and arguments presented at trial, we find the preponderance of the evidence shows that the ’317 Application Specification fails to provide adequate written description support for independent claims 1 and 8. As such, none of the Challenged Claims are entitled to claim the benefit of the filing date of any of the priority applications. The effective filing date of the Challenged Claims is the actual filing date of the ’803 Patent, which is July 10, 2020, thereby making the claims eligible for post-grant review.

IV. ANALYSIS OF GROUNDS

To prevail in this post-grant review of the challenged claims, Petitioner must prove unpatentability by a preponderance of the evidence. 35 U.S.C. § 326(e); 37 C.F.R. § 42.1(d). The petitioner has the burden from the onset to show with particularity why the challenged claims are unpatentable. *See Harmonic Inc. v. Avid Tech., Inc.*, 815 F.3d 1356, 1363 (Fed. Cir. 2016). This burden of persuasion never shifts to the patent owner. *See Dynamic Drinkware, LLC v. Nat'l Graphics, Inc.*, 800 F.3d 1375, 1378 (Fed. Cir. 2015) (discussing the burden of proof in *inter partes* review).

A. Ground 1: No Written Description

Petitioner asserts that claims 1–8, 12 and 20 of the '803 Patent are unpatentable for lack of written description for the same reasons set forth above. Pet. 96.

Having considered the arguments and evidence presented at trial, because the '803 Patent Specification is substantively identical to the priority application discussed above, we determine Petitioner has also shown by a preponderance of the evidence that the Challenged Claims are unpatentable for lack of written description for the same reasons discussed above with respect to priority.

B. Remaining Grounds

In the remaining challenges set forth in the Petition, Petitioner asserts claims 1–8, 12, and 20 of the '803 Patent are unpatentable for lack of enablement (Ground 2), are anticipated by Byrd (Ground 3) and obvious over Byrd and IMBRUVICA Label (Ground 4).

Because we have determined that all of the challenged claims are unpatentable for lack of written description, which is dispositive as to all Challenged Claims, we need not reach these remaining grounds. *See SAS*

Inst. Inc. v. Iancu, 138 S. Ct. 1348, 1359 (2018) (holding that a petitioner "is entitled to a final written decision addressing all of the claims it has challenged"); *Bos. Sci. Scimed, Inc. v. Cook Grp. Inc.*, 809 F. App'x 984, 990 (Fed. Cir. 2020) (non-precedential) (recognizing that the "Board need not address issues that are not necessary to the resolution of the proceeding" and, thus, agreeing that the Board has "discretion to decline to decide additional instituted grounds once the petitioner has prevailed on all its challenged claims").

V. CONCLUSION[23]

For the foregoing reasons, we determine that Petitioner has established by a preponderance of the evidence that claims 1–8, 12, and 20 of the ’803 Patent are unpatentable.

In summary:

Claim(s)	**35 U.S.C. §**	**Reference(s)/Basis**	**Claim(s) Shown Unpatentable**	**Claim(s) Not shown Unpatentable**
1–8, 12, 20	112	Written Description	1–8, 12, 20	
1–8, 12, 20	112	Enablement[24]		
1–8, 12, 20	102	Byrd[25]		
1–8, 12, 20	103	Byrd, IMBRUVICA Label[26]		
Overall Outcome			1–8, 12, 20	

[23] Should Patent Owner wish to pursue amendment of the challenged claims in a reissue or reexamination proceeding subsequent to the issuance of this decision, we draw Patent Owner’s attention to the April 2019 *Notice Regarding Options for Amendments by Patent Owner Through Reissue or Reexamination During a Pending AIA Trial Proceeding*. *See* 84 Fed. Reg. 16,654 (Apr. 22, 2019). If Patent Owner chooses to file a reissue application or a request for reexamination of the challenged patent, we remind Patent Owner of its continuing obligation to notify the Board of any such related matters in updated mandatory notices. *See* 37 C.F.R. § 42.8(a)(3), (b)(2).

[24] Because all challenged claims have been found unpatentable for lack of written description, we do not address this ground.

[25] Because all challenged claims have been found unpatentable for lack of written description, we do not address this ground.

[26] Because all challenged claims have been found unpatentable for lack of written description, we do not address this ground.

VI. ORDER

In consideration of the foregoing, it is hereby:

ORDERED that claims 1–8, 12, and 20 of U.S. Patent No. 11,672,803 B2 are held unpatentable; and

FURTHER ORDERED that, because this is a Final Written Decision, the parties to the proceeding seeking judicial review of the decision must comply with the notice and service requirement of 37 C.F.R. § 90.2.

PETITIONER:

David Berl
WILLIAMS & CONNOLLY LLP
dberl@wc.com

PATENT OWNER:

David Frazier
Michael Morin
Brenda Danek
LATHAM & WATKINS LLP
david.frazier@lw.com
michael.morin@lw.com
brenda.danek@lw.com